

Automated Soil Slope Classification for Statewide GIS Modeling

An ArcGIS Python Toolbox for downloading, validating, and classifying USDA soil survey data across Ohio's 88 counties

The Problem

Land-use models that classify terrain (e.g., for transportation or infrastructure planning) need accurate slope data derived from soil survey information. That data comes from the USDA's SSURGO (Soil Survey Geographic Database), county by county, and getting it into a usable state is normally a slow, manual, and error-prone process:

- Publication dates for each county's soil data change over time, so URLs can't be hardcoded — a stale link either 404s or silently returns an outdated dataset.
- Downloads can fail partway through or return HTML error pages instead of real data, and nothing catches that until it breaks a downstream step.
- A statewide run touches 88 separate counties; if it fails on county 60, there's no way to resume without starting over.
- The raw output needs to be classified into meaningful terrain categories (Level / Moderate / Steep) and checked for data quality before anyone downstream can trust it.

The tool needed to turn a manual, single-county workflow into a reliable, auditable, statewide (or district- or county-scoped) batch process that a GIS analyst could kick off and walk away from.

My Approach

I built this as a custom **ArcGIS Python Toolbox (.pyt)**, so it plugs directly into ArcGIS Pro as a native geoprocessing tool with a proper parameter UI — no separate script execution required.

Dynamic data sourcing, not hardcoded assumptions. Rather than pointing at fixed download URLs, the tool queries USDA's Soil Data Access API at runtime to get each county's current publication date, then builds the download URL from that. If the API is unreachable, it clearly warns the operator instead of failing silently.

Defense in depth on downloads. Every downloaded zip is checked for a minimum file size and a valid zip header before being trusted — catching the common failure mode where a server returns a small HTML error page with a `.zip` extension.

Two-tier fallback for slope data. Slope values are pulled directly from the SDA API first (faster, no need to parse tabular files), with automatic fallback to parsing the raw

`muaggatt.txt` file from the extracted survey package if the API is unavailable. Either path feeds the same classification logic.

Checkpointing for a statewide run. Processing 88 counties takes time and can fail midway (network blip, bad data, etc). Rather than treating this as all-or-nothing, the tool skips counties that already have output on disk (with an explicit override flag), so a failed statewide run can simply be re-launched and pick up where it left off.

Configurable, documented classification thresholds. Slope is classified into Level / Moderate / Steep bands against thresholds that are exposed as tool parameters (defaulting to project-standard values) rather than buried as magic numbers, so analysts can adjust them without touching code.

Built-in QA, not just output. Each county's result is automatically flagged (`PASS`), (`WARN_HIGH_NODATA`), (`WARN_LOW_COVERAGE`), (`SKIPPED`), (`FAILED`) based on data coverage thresholds, so quality issues surface immediately instead of being discovered downstream. The tool also validates its own prerequisites (e.g., that the required base layer exists and is populated) before doing any work, so failures produce a clear message instead of a cryptic stack trace.

Structured, dual-format reporting. Every run produces:

- A pipe-delimited QAQC report (per-county timing, data sources used, feature counts, slope statistics, QA flags) — formatted for both human review and direct database import.
- A detailed CSV at the map-unit level, suitable for personal audit or downstream analysis.

Technical Highlights

- **Language/Platform:** Python, ArcGIS Pro (`arcpy`) Python Toolbox
- **Integration:** USDA Soil Data Access REST API (JSON queries), USDA Web Soil Survey download service
- **Data handling:** Shapefile processing, cursor-based field updates, feature class merging, geodatabase I/O
- **Reliability engineering:** API-first with file-based fallback, download integrity validation, resumable batch processing, upfront input validation
- **Scope flexibility:** Supports statewide, ODOT-district, and single-county runs from the same tool via a built-in county-to-district mapping for all 88 Ohio counties
- **Reporting:** Auto-generated QAQC logs and CSV exports designed for both human review and machine (database) ingestion

Why It Matters

This tool turns a data-acquisition step that would otherwise require a person manually tracking 88 counties' worth of download links, file integrity, and slope classification into a single, self-auditing, restartable geoprocessing run. The QA flagging and detailed logging mean data quality issues are caught at the source rather than discovered several pipeline stages later — which matters a lot when the output feeds a statewide land-use model that other decisions depend on.